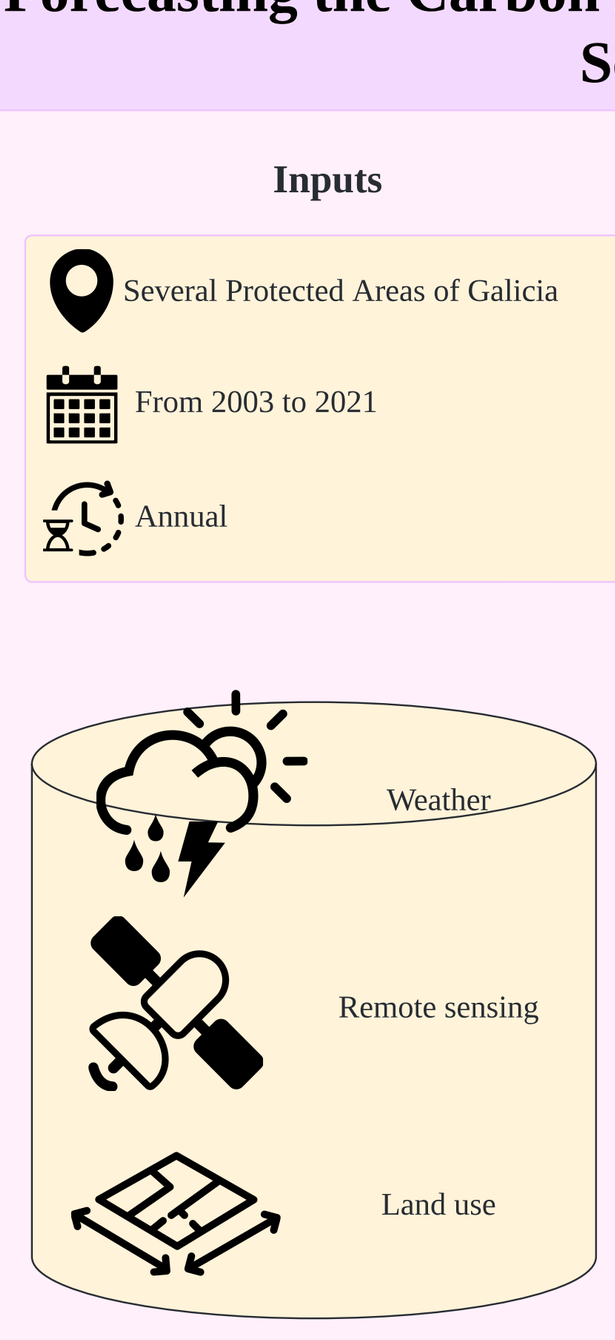
Forecasting the Carbon Absorption of Protected Areas Under Future Climatic and Land Use Scenarios Using Machine Learning Algorithms

Inputs

Several Protected Areas of Galicia

From 2003 to 2021

Annual



RF:
8% of error
80% of correlation

GBT:
5% of error
91% of correlation

MLPNN:
7% of error
86% of correlation

Training

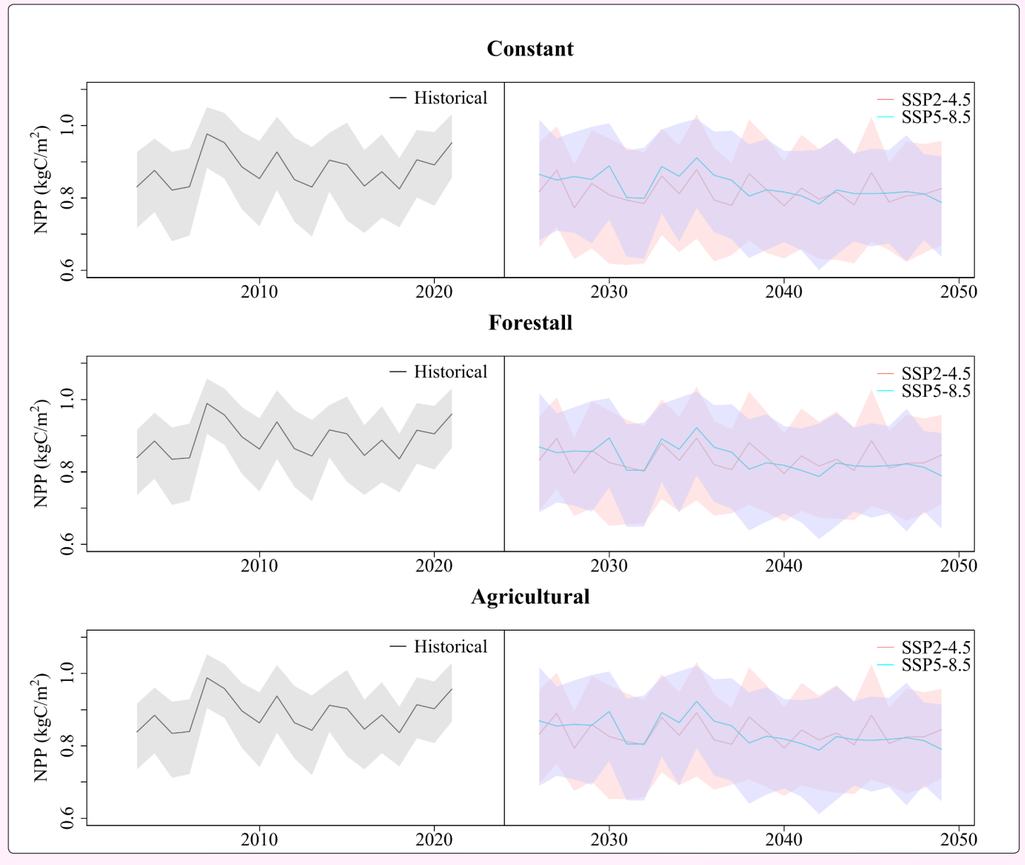
RF:
7% of error
88% of correlation

GBT:
4% of error
95% of correlation

MLPNN:
9% of error
77% of correlation

Testing

Results: Land use scenario forecast



Conclusion

A reduction on the average carbon absorption of the Protected Areas of Galicia is expected for the near future no matter the climatic projection. The decreasing trend for the long-term future, nonetheless, seems to be significant only for the fosil fuel-based developement.

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